Digital Twin Data Management: A Comprehensive Review

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(Survey-Tutorial Paper)

***Abstract*—Digital Twins are virtual representations of physical assets and systems that rely on effective Data Management to integrate, process, and analyze diverse data sources. This article comprehensively examines Data Management challenges, architectures, techniques, and applications in the context of Digital Twins. It explores key issues such as data heterogeneity, quality assurance, scalability, security, and interoperability. The paper outlines architectural approaches like centralized, distributed, cloud-based, and blockchain solutions and Data Management techniques for modeling, integration, fusion, quality management, and visualization. Domain-specific considerations across manufacturing, smart cities, healthcare, and other sectors are discussed. Finally, open research challenges related to standards, real-time data processing, intelligent Data Management, and ethical aspects are highlighted. By synthesizing the state-of-the-art, this review serves as a valuable reference for developing robust Data Management strategies that enable Digital Twin deployments.**

***Index Terms*—Digital twin, data management, data integration, data quality, data security, data visualization.**

I. Introduction

A DIGITAL Twin (DT) is a virtual representation of a physical object or system that spans its entire lifecycle, employing real-time data and other available data to enable learning, reasoning, and ultimately decision-making [1], [2]. DTs integrate artificial intelligence, machine learning, and software analytics with data from multiple sources to create living digital simulation models that update and change as the physical counterpart changes [3], [4]. The concept of a DT originates from NASA's endeavors to create ultra-realistic simulation models of space vehicles and systems [5], [6]. However, emerging technologies like the Internet of Things (IoT), advanced sensors, Big Data analytics, and cloud computing have enabled DTs to proliferate across various industries [7], [8], [9], [10]. At its core, a DT serves as the digital counterpart of a physical asset, process, system, or even an entire environment such as a city or factory [11], [12], [13], [14]. It continuously updates and mirrors the life of its physical twin through the seamless exchange and incorporation of data flows from the physical world [2], [3], [15]. This bidirectional interaction between the virtual and physical worlds enables real-time monitoring, simulation, optimization, prediction, and control of the physical asset or system [1], [5], [16]. The potential benefits of DTs span improved system understanding, optimized performance, predictive maintenance, faster decision-making, cost reduction, and enhanced safety and reliability [1], [5], [17]. As such, DTs are increasingly finding applications across diverse domains like manufacturing, construction, healthcare, smart cities, energy, transportation, and agriculture [10], [11], [18], [19], [20], [21], [22].

The successful deployment and operation of DTs hinge on effective Data Management (DM) strategies that can handle the massive volumes of multi-source, multi-modal data generated by these virtual replicas [23], [24], [25], [26]. DM encompasses the acquisition, integration, processing, analysis, and governance of data throughout its lifecycle to create accurate, up-to-date virtual models that mirror the physical counterpart [12], [27], [28]. The challenges associated with DM in DTs are multifaceted, ranging from data heterogeneity, quality assurance, security, and privacy to scalability, performance, and interoperability [23], [29], [30], [31], [32], [33].

A. Motivations

The emergence of DTs as virtual representations of physical entities has profound implications for various industries, revolutionizing how complex systems are designed, operated, and optimized. At the core of this transformative paradigm lies the critical task of managing and harnessing the massive volumes of data generated by these dynamic virtual replicas. The success of the DT concept hinges on our ability to effectively manage and extract value from the data deluge brought about by the Industry

Received 6 April 2024; revised 6 January 2025; accepted 20 January 2025. Date of publication 27 January 2025; date of current version 4 September 2025. Recommended for acceptance by L. Tang. *(Corresponding authors: Ammar Hawbani and Xingfu Wang.)*

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Digital Object Identifier 10.1109/TBDATA.2025.3533891

4.0 and IoT revolution. As enterprises increasingly rely on DTs to optimize operations, anticipate maintenance needs, and enhance decision-making, the quantity, velocity, and diversity of data they produce have escalated dramatically. Consequently, the management of this data, its interpretation, and the extraction of actionable insights have emerged as paramount challenges that must be addressed to unlock the full potential of DTs. This study is motivated by the realization that effective DM is a fundamental enabler for the successful deployment and operation of DTs. The overarching question that arises is: How can we effectively handle the diverse and dynamic data streams within DT ecosystems?

Our primary motivations for undertaking this comprehensive review of DM for DT Systems are threefold:

1) *Bridging the Knowledge Gap:* While the value proposition of DTs is widely acknowledged, there exists a knowledge gap regarding the intricate details of DM within these virtual counterparts. This study aims to fill this void by providing an in-depth analysis of the current landscape and best practices.
2) *Accelerating DT Adoption:* We believe that a thorough understanding of effective DM techniques can catalyze the adoption of DTs across industries. By showcasing successful deployments and best practices, this paper endeavors to pave the way for organizations seeking to leverage the power of DTs.
3) *Shaping Future Research Directions:* The opportunities and challenges posed by effective DM in DT present fertile ground for future research endeavors. Through this analysis, we seek to highlight promising areas for further exploration and encourage academics to investigate innovative solutions in this rapidly evolving domain.

B. Contributions

This comprehensive survey makes several key contributions that provide a holistic perspective on data management for digital twins. We systematically examine the challenges, architectures, techniques, and application domains, offering a valuable resource for researchers, practitioners, and organizations seeking to develop effective data management strategies. The primary contributions of this study are as follows:

1) A detailed analysis of the critical data management challenges faced by digital twin systems, including data heterogeneity, quality assurance, scalability demands, security and privacy concerns, and governance complexities. This comprehensive overview lays the foundation for developing robust data management strategies tailored to the unique requirements of digital twins.
2) An in-depth exploration of architectural paradigms and frameworks for data management in digital twin environments, encompassing centralized, distributed, cloud-based, and blockchain-enabled approaches. The paper evaluates the strengths, limitations, and trade-offs of these architectures, providing a guiding framework for organizations to select the most suitable approach based on their specific needs and constraints.
3) A thorough examination of key data management techniques and technologies essential for robust digital twin implementations. The paper delves into aspects such as data modeling and ontologies, data fusion and integration, data quality management, secure access control mechanisms, and innovative data visualization approaches. This comprehensive coverage equips researchers and practitioners with a solid foundation for developing cutting-edge data management solutions.
4) An extensive exploration of data management considerations and best practices across various application domains where digital twins are being deployed, including manufacturing, smart cities, construction, healthcare, energy, transportation, and agriculture. This domain-specific analysis provides valuable insights into the unique data management challenges and opportunities within each sector, facilitating the effective translation of digital twin technologies across industries.

C. Scope and Organization

This article provides a comprehensive overview of DM challenges, architectures, techniques, and applications in the domain of DTs. The scope covers DM aspects across the entire lifecycle of DTs, from data acquisition and modeling to integration, processing, analysis, visualization, and governance. The paper aims to serve as a comprehensive reference for researchers and practitioners working on developing robust DM solutions for DT systems across diverse domains. It highlights the state-of-the-art, identifies gaps, and outlines promising future research directions in this critical area.

The rest of this article is organized as follows. Section III outlines the key DM challenges faced in developing and operating DTs effectively. It discusses issues related to data heterogeneity, quality, security, privacy, scalability, and performance requirements. Section IV reviews different architectural approaches and frameworks for DM in DTs. This includes centralized, distributed, cloud-based, and blockchain-based DM architectures. Section V delves into the various DM techniques and technologies relevant to DTs. Topics covered include data modeling and ontologies, data fusion and integration, data quality management, data security, and access control mechanisms, and data visualization and representation techniques. Section VI explores DM considerations and approaches in different application domains where DTs are being deployed, such as manufacturing, smart cities, construction, healthcare, energy, transportation, and agriculture. Section VII discusses open research challenges, opportunities, and future directions in DM for DTs. This includes aspects like interoperability, real-time data processing, intelligent DM, and ethical and regulatory issues. Finally, Section VIII provides a concise conclusion summarizing the key points covered in the article.

II. Digital Twin and Data Management

A. Digital Twin: Concepts and Definitions

Various definitions of DT have emerged in recent years, as reviewed by Tao et al. [1] and Negri et al. [34]. Currently,

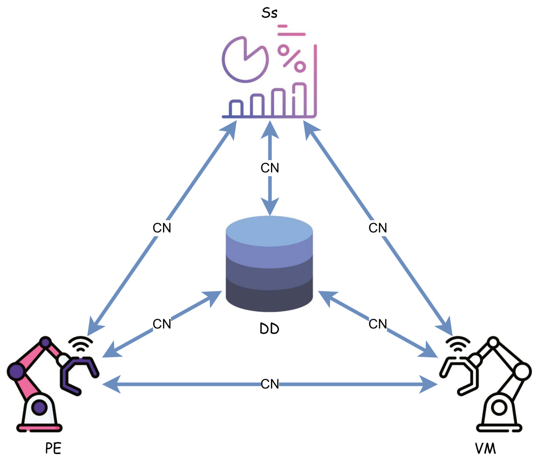

Fig. 1. Five-dimension digital twin model.

the two most widely accepted definitions were provided by Grieves and NASA. NASA defined DT for a space vehicle as "A Digital Twin is an integrated multi-physics, multiscale, probabilistic simulation of an as-built vehicle or system that uses the best available physical models, sensor updates, fleet history, etc., to mirror the life of its corresponding flying twin" [35]. In 2014, Grieves published a white paper on DT, proposing a basic model consisting of three main parts: (a) physical products in Real Space, (b) virtual products in Virtual Space, and (c) the connections of data and information that ties the virtual and real products together [36]. The three-dimension DT model originally defined by Grieves has been widely applied. However, with the expansion of DT applications into civilian fields and the increasing complexity of service demands, Tao et al. [37] proposed a five-dimension DT model to promote further applications in diverse fields. This model can be formulated as:

$$DT = \{PE, VM, Ss, DD, CN\} \tag{1}$$

where PE are physical entities, VM are virtual models, Ss are services, DD is DT data, and CN are connections. The five-dimension DT model is illustrated in Fig. 1.

From the proposed models, it is evident that data plays a central role in the digital twin ecosystem. Twin data is recognized as a key driver of DT [38]. The data in a DT system is characterized by its multi-temporal scale, multi-dimension, multi-source, and heterogeneous nature. Various types of data contribute to the DT ecosystem, including static attribute data, dynamic condition data, simulation results, service invocation and execution data, knowledge from domain experts, and fusion data resulting from the integration of all these data types. The connections between physical entities, virtual models, services, and data enable information and data exchange, facilitating advanced simulation, operation, and analysis.

B. Importance of Data Management for Digital Twins

Effective DM lies at the heart of realizing the full potential of DTs. DTs are data-driven virtual representations that rely on the seamless acquisition, integration, processing, and analysis of massive volumes of multi-modal, multi-source, heterogeneous data from a variety of sources such as sensors, control systems, enterprise systems, external data feeds, and human inputs [23], [24], [25], [26], [29]. This data needs to be systematically managed throughout its lifecycle to create an accurate, up-to-date virtual model that can effectively mirror the current state and predict the future behavior of the physical counterpart [12], [27], [28]. Some key reasons why robust DM is critically important for DTs include:

1) Data Variety and Complexity: DTs ingest and fuse a wide variety of structured, unstructured, and semi-structured data formats from multiple domains at different velocities. Managing this data heterogeneity, providing semantic consistency, and enabling meaningful data integration across silos is a fundamental challenge [29], [33], [39], [40], [41], [42].

2) Data Quality and Trustworthiness: The accuracy, reliability, and trustworthiness of insights derived from DTs are heavily dependent on the quality of the input data. Ensuring data quality through profiling, cleansing, and rigorous governance processes is essential to generate reliable predictions and decisions [7], [23], [30], [43].

3) Data Security and Privacy: DTs often deal with sensitive data related to products, operations, systems, and individuals. Implementing robust data security measures, access controls, and ensuring regulatory compliance with data privacy laws is of paramount importance [31], [44], [45], [46].

4) Data Scalability and Performance: The data volumes, velocity, and complexity involved with industrial and urban-scale DTs can be massive. Highly scalable and high-performance DM architectures and systems are needed to handle the data deluge in a timely manner [29], [32].

5) Data Interoperability and Integration: Data for DTs comes from diverse heterogeneous sources and must be semantically integrated. Addressing data interoperability issues, developing data standards, and ensuring seamless data exchange across systems are key challenges [2], [33], [47], [48], [49].

6) Data Visualization and Analytics: Intuitive visualization of the large, multi-dimensional DT data and effective analytics techniques are required to derive valuable insights from the virtual models to support decision-making [16], [25], [50]. Robust DM architectures, models, and technologies that can effectively address these multifaceted challenges are essential to truly unlock the transformative potential of DTs across various industries [51], [52]. This comprehensive review highlights the state-of-the-art and open research challenges in DM for DTs.

C. Related Studies

The concept of digital twins has gained significant attention in recent years, with various literature reviews exploring different aspects of this technology. Early reviews, such as Kritzinger et al. [53] in 2018, provided a general overview of digital twins in manufacturing, while Negri et al. [34] in 2017 examined the roles of digital twins in cyber-physical production systems. However, these initial reviews did not specifically focus on data management challenges and solutions. As the field progressed, researchers began to acknowledge the importance of effective data management for digital twin implementations. Semeraro et al. [6] in 2021 conducted a comprehensive review of digital twin technology, recognizing data management as a crucial aspect but without delving into its specifics. In 2020, Jones et al. [15] and Liu et al. [54] published dedicated reviews characterizing digital twins and examining their concepts, technologies, and industrial applications. While these reviews touched upon data management considerations, they did not provide an in-depth analysis of data management challenges, architectures, and techniques specific to digital twins. Very few reviews have explicitly addressed data management in digital twins, albeit with varying scopes and depths. Correia et al. [23] in 2023 conducted a systematic literature review on data management in digital twins, highlighting key challenges and solutions. Additionally, some domain-specific reviews have explored data management aspects within their respective sectors. For instance, Opoku et al. [55] in 2021 reviewed digital twin applications in the construction industry, which inherently involved data management considerations. While these existing reviews have contributed valuable insights, there is a need for a comprehensive and up-to-date review that synthesizes the state-of-the-art in data management for digital twins across various domains and applications. This paper aims to fill this gap by providing a holistic overview of data management challenges, architectures, techniques, and applications in the context of digital twins, as well as highlighting open research challenges and future directions.

III. Data Management Challenges

DTs rely on the continuous acquisition, integration, and processing of massive amounts of multi-source, multi-modal data to create accurate virtual representations of physical assets, systems, or environments. Effectively managing this data deluge poses several formidable challenges that must be addressed to realize the full potential of DTs. This section outlines some of the key DM challenges associated with DT systems.

The heterogeneous nature of data constitutes a fundamental challenge. DTs ingest data from a wide variety of sources such as sensors, control systems, enterprise systems, human inputs, and external data feeds in numerous formats (structured, semi-structured, unstructured) and models [40], [41]. Providing semantic integration and interoperability across this data variety is critical [29], [33]. Data quality is another crucial aspect, as insights from DTs are only as reliable as the input data [23], [30], [43]. Issues like missing data, noise, outliers, and data drift can significantly impact analytic accuracy. Robust data quality assurance, cleansing and governance processes are essential. The scale and performance requirements of DT DM can be extremely demanding, especially for industrial and urban-scale deployments [32]. The ability to ingest, process and analyze high-velocity, high-volume data streams in real-time or near real-time is vital [29]. Data security and privacy are top priorities when dealing with sensitive operational and customer data [31], [45]. Robust cybersecurity measures, access controls, encryption, and privacy-preserving DM, are necessary [44]. As DTs evolve over time, the data models representing them also need to change [39]. Data evolution management to dynamically update and version data models while maintaining consistency is an important challenge. Additionally, DTs often involve collaboration between multiple stakeholders, necessitating secure data sharing, distributed governance, and traceability [31], [46].

In the subsequent sections, we delve deeper into these key DM challenges faced by DT systems across different dimensions. Effective strategies to overcome these obstacles are critical for unleashing the true potential of DTs.

A. Data Heterogeneity and Integration

DTs are characterized by a convergence of diverse data streams from a multitude of sources, formats, modalities, and semantic models across different domains [7], [24], [41]. Integrating and providing a unified view of this extremely heterogeneous data landscape poses a significant challenge [33]. On one hand, DTs must ingest and consolidate data arising from the physical counterpart itself, including sensor data captured by a myriad of IoT devices, machines, equipment, and facilities [7], [24]. This data can arrive in various structured formats like relational databases or time-series datasets, semi-structured formats like XML or JSON, as well as unstructured formats such as multimedia files, documents, and human-recorded inputs [41]. Furthermore, DTs also need to integrate design and simulation data represented using standards and tools like computer-aided design (CAD), building information modeling (BIM), finite element method (FEM), and other modeling paradigms [8], [11], [18]. This data is essential for creating the as-designed virtual models that form the basis of the DT. However, merging this data with the as-operated sensor data streams presents semantic integration challenges. On top of this, DTs must also incorporate relevant enterprise data residing in systems like enterprise resource planning (ERP), product lifecycle management (PLM), manufacturing execution systems (MES), and other operational data stores [28], [41], [47]. Each of these systems often follows its own data models, conventions, and schemas. The data integration complexities are further exacerbated by the need to fuse DT data with relevant external data contexts such as weather data, geospatial data, social media data, and other third-party feeds [22], [56]. These external data sources bring their own formats, velocities, and semantic representations into the mix. Resolving the semantic heterogeneities and providing meaningful data unification across this diverse data ecosystem is an enormous challenge that currently relies heavily on manual, domain-specific data mapping and transformation rules [23].

While techniques like ontology mapping, schema integration, entity resolution, data cleansing and transformation can aid in data integration, they face issues of scalability and often require significant human effort [39], [40], [57]. Automated approaches based on machine learning, Semantic Web technologies, and knowledge graphs show promise but have their own limitations [40], [57]. Developing efficient, scalable and robust data integration pipelines that can handle the full scope and variety of DT data remains an open issue. The lack of common data standards and models across DT applications further exacerbates these integration complexities [2], [47], [48], [49].

B. Data Quality and Reliability

The insights, predictions, and decisions derived from a DT are only as accurate and reliable as the input data streams feeding the virtual model. However, ensuring high data quality and reliability poses a formidable challenge given the diverse nature of data sources, modalities, and dynamic operational conditions involved in DTs. Data quality issues can arise from various factors - sensor failures or calibration errors can lead to missing, noisy, or erroneous data [30], [43]. Network connectivity issues, human errors during data entry, and race conditions can also result in incomplete, duplicate, or inconsistent data across sources [23], [30]. The data collected from the physical environment may also contain outliers, anomalies, or artifacts that do not accurately represent the true operating conditions [30]. Furthermore, the data distributions and patterns can shift over time due to concept drift as operating contexts and conditions evolve, rendering historical data obsolete [43]. Such data quality deficiencies can severely undermine the accuracy of the DT's virtual models, leading to faulty system understanding, flawed predictions, and unreliable decision-making outcomes [23], [30], [43].

To address these challenges, DTs require robust data quality assurance processes involving techniques like data validation, cleansing, outlier removal, interpolation to impute missing values, and automated error detection and correction routines [23], [58]. However, designing unsupervised data cleaning methods that can handle the scale, velocity, and multi-modality of DT data streams in a scalable and adaptive manner remains an open challenge. Continuous monitoring of data quality metrics and proactive detection of data drift patterns is also crucial to maintain reliability as operational conditions change over the DT's lifecycle [30], [43]. Cutting-edge machine learning approaches like reinforcement learning, transfer learning, and adversarial training show promise in developing more intelligent and adaptive data quality management solutions [59], [60], [61]. However, their effectiveness in practical DT scenarios with limited labeled data remains to be evaluated [62]. From a systems perspective, data quality governance through standardized processes, rigorous metadata management, master DM, and data lineage tracking are essential [40], [49]. This is especially critical in multi-stakeholder DT environments where data sovereignty and regulatory compliance considerations come into play. Ultimately, beyond just monitoring and correcting data quality issues, robust mechanisms to quantify uncertainties inherent in DT data and principled techniques to model these uncertainties in virtual representations are vital for ensuring the trustworthiness and reliability of insights [5], [27], [63]. Probabilistic modeling, imprecise computation and uncertainty quantification methods from sciences like meteorology could potentially offer avenues to address these challenges.

C. Data Security and Privacy

DTs, by their very nature, deal with large volumes of sensitive data related to physical assets, operational processes, intellectual property, and even individuals. Ensuring the security and privacy of this data is therefore of paramount importance, yet it poses significant challenges. On the security front, DT systems are vulnerable to cyber threats like unauthorized access, data breaches, malicious data injection attacks, and adversarial machine learning attacks [31], [45]. The consequences of such attacks can be severe, ranging from intellectual property theft, operational disruptions, and safety incidents to regulatory penalties. The distributed and multi-source nature of DT data exacerbates these security risks. Data flows from diverse sources like edge devices, enterprise systems, cloud platforms, and third-party data feeds, creating a broader attack surface. Implementing robust cybersecurity controls, access management protocols, data encryption, and secure communication channels across this fragmented ecosystem is extremely challenging [31], [46]. Traditional cybersecurity approaches often fall short in addressing the unique requirements of DT environments. Data privacy is another critical aspect, as DTs may process personal data related to employees, customers, or citizens depending on the application domain. Ensuring compliance with data privacy regulations like GDPR, CCPA, and HIPAA while supporting DT operations is a delicate balancing act [45]. Techniques like data anonymization, differential privacy, and federated learning show promise but have limitations in terms of utility preservation and scalability [44], [45]. Moreover, DTs often involve multi-party data sharing and collaboration scenarios where data sovereignty, governance, and trust issues come into play [31], [46], [64]. Blockchain and distributed ledger technologies have been proposed as potential solutions to enable secure, trusted, and auditable data sharing without compromising privacy and intellectual property [7], [31], [45], [46], [65], [66]. However, their scalability, performance overheads, and integration challenges in enterprise DT environments remain open research questions.

From a systems perspective, robust data security and privacy-preserving measures need to be baked into the architectural design of DT platforms and implemented in a holistic manner across the entire data lifecycle [31], [45]. This includes secure data acquisition, storage, transmission, processing, and controlled access mechanisms. Developing comprehensive risk assessment frameworks and cyber-resilient DT architectures resilient to attacks and failures is also crucial, especially for mission-critical applications [31]. Ultimately, given the high-stakes nature of DT data, a careful balancing of security, privacy, trust, and utility preservation requirements is essential. Continuous research into novel cryptographic techniques, secure computing paradigms, and policy frameworks is needed to address these multi-faceted challenges.

D. Data Scalability and Performance

The scale and velocity of data involved in DT systems, especially for industrial and urban-scale deployments, can be truly staggering. DTs must be able to ingest, process, and analyze massive streams of real-time data flowing in from thousands or even millions of sensors, machines, and IoT devices [29], [32]. This data deluge is further compounded by the need to fuse live operational data with historical data, simulated data, and diverse external data feeds in order to construct an accurate, up-to-date virtual model. Handling such high volumes of multi-modal data streams in a timely and responsive manner is a formidable DM

TABLE I
COMPARISON OF DATA MANAGEMENT ARCHITECTURES

Architecture	Advantages	Disadvantages	Key Characteristics
Centralized [67]–[69]	• Tight control & governance • Data consolidation • Simplified integration & processing	• Single point of failure • Scalability limitations • Data sovereignty issues	• Central data repository/lake • End-to-end data pipeline • Holistic analytics & visualization
Distributed [29], [31], [70]	• Process data near sources • Low latency • Improved resilience • Data sovereignty	• Data consistency challenges • Complex data flows • System orchestration overhead	• Edge/on-premise nodes • Data virtualization • Federated queries • Aligns with IDS, IIRA concepts
Hybrid [31], [58], [65], [70]	• Balances centralized & distributed strengths • Supports real-time edge processing • Centralized governance & analytics	Complexity of hybrid data flows • Data consistency across tiers	• Edge preprocessing • Central data integration • Handles scale & stakeholder complexity
Cloud-based [32], [71], [72]	• Scalability & elasticity • Global accessibility • Integrated data services	• Data sovereignty concerns • Vendor lock-in risks • Data transfer overheads	• Cloud data lakes/warehouses • Serverless data processing • IoT/DT cloud platforms
Blockchain-based [7], [45], [46], [65]	• Decentralization • Data immutability • Enhanced trust & provenance • Secure data sharing	Scalability & throughput limits • Standardization challenges • Consortium governance complexity	• Distributed ledger • Smart contracts • Respects data sovereignty • Multi-party integration

challenge. The data pipelines must be able to perform complex transformations, cleansing, integration, and analytics on this torrent of data with minimal latencies to enable real-time or near real-time monitoring, prediction, and control capabilities expected of DTs [32], [58]. Any delays or performance bottlenecks can potentially undermine the effectiveness of the DT, especially in mission-critical applications with strict reliability and responsiveness requirements. The scalability demands are not just limited to data ingestion and processing but also extend to other aspects like data storage, retrieval, and visualization. As DTs continuously generate new data over their lifecycle, the storage and DM infrastructure must be able to scale elastically to accommodate this growth without compromising performance [65]. Efficiently storing, indexing, and retrieving historical data for tasks like anomaly detection, root cause analysis, and what-if scenario exploration is crucial. Data visualization is another key aspect that requires scalable solutions, as DTs often need to render large, high-dimensional datasets in intuitive and interactive visual formats for human comprehension and decision support [16], [25], [50]. Transforming the raw data into insightful visualizations in a responsive manner while handling the dynamism and complexity of the data poses significant engineering challenges.

From a systems perspective, technologies like distributed computing frameworks, in-memory data grids, GPUs and hardware accelerators have become essential to meet the extreme scalability and performance demands of DT DM [32]. Cloud computing platforms, with their elastic scaling capabilities and serverless architectures, are also increasingly being leveraged [32]. However, managing data distribution, consistency, and governance in such distributed data infrastructures is nontrivial. Developing DM solutions that can dynamically adapt and scale resources in response to fluctuating workloads while optimizing for cost and energy efficiency is an active research area [29], [32]. Edge computing paradigms that can offload certain data processing tasks closer to the source are also gaining traction [31]. Ultimately, a holistic approach that harmonizes edge, cloud, and on-premise infrastructures may be required for scalable, high-performance DT DM.

IV. Data Management Architectures and Frameworks

DM is a critical enabler for realizing the potential of DTs, underpinning functions like data integration, processing, analytics, and governance. Several architectural paradigms and frameworks are available for managing the diverse and complex data landscapes associated with DTs. The choice of architecture depends on factors such as the scale of deployment, distribution of data sources, performance requirements, and stakeholder ecosystem complexity. Table I compares the key attributes of different DM architectures explored in this section - centralized, distributed, hybrid, cloud-based, and blockchain-based approaches. Each architecture presents unique advantages and trade-offs in terms of scalability, control, resilience, data sovereignty, and trusted data-sharing capabilities for DT implementations. As DT adoption accelerates across industries, innovative DM architectures that harmoniously blend these paradigms are emerging. Hybrid models that combine centralized data governance with distributed processing at the edge, augmented by cloud elasticity and secure blockchain integration, can holistically address the multifaceted DM needs of large-scale DT ecosystems. The following subsections explore each of these architectural approaches in detail.

A. Centralized Architecture

One of the most traditional and widely adopted approaches for DM in DTs is the centralized architecture. In this paradigm, all the data from various sources like sensors, enterprise systems, external feeds, etc., is collected and stored in a central repository or data lake [67], [68], [69] as shown in Fig. 2. The core DM functions such as integration, processing, analytics, and visualization are also performed in this centralized system [73]. The key advantage of this approach is that it allows for tight control, governance, and security over the complete DT data pipeline within a unified platform [67]. Data silos are eliminated as all relevant data converges into a single environment. This centralized data consolidation also simplifies tasks like master DM, metadata management, and enforcing data quality rules. Many commercial and open-source platforms tailored

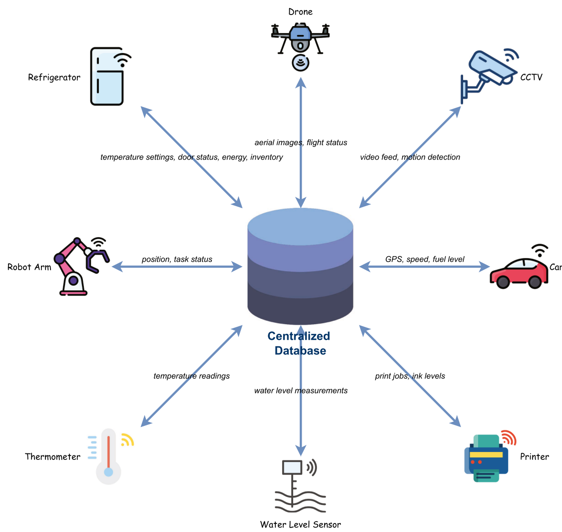

Fig. 2. Centralized architecture.

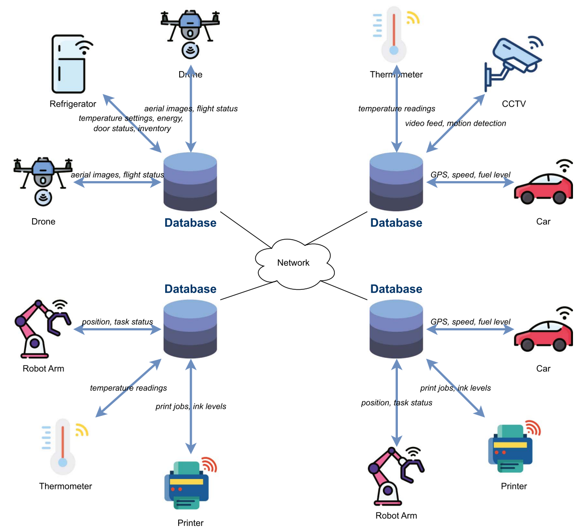

Fig. 3. Distributed architecture.

for Industrial IoT, Industry 4.0, and smart manufacturing use centralized DM architectures for DTs [26]. Examples include Siemens MindSphere, PTC Thingworx, Dassault 3DExperience, and open platforms like FIWARE [74]. These typically provide end-to-end capabilities from data ingestion, storage, and integration to analytics and visualization tailored for industrial DTs. Centralized architectures have also been applied in other domains like construction, where integrated data environments consolidate building information models (BIMs), sensor data, and project data for building DTs [8], [11], [55], [75]. Platforms like Autodesk Forge employ centralized data lakes. Similarly, for smart city DTs, centralized data platforms help unify IoT, geospatial, and other city data sources [13], [14].

However, centralized DM faces challenges with scalability, single point of failure risks, lack of interoperability, and data sovereignty issues in multi-stakeholder scenarios [31]. It can also be challenging to consolidate and process large-scale streaming data from distributed sources in a centralized fashion with acceptable latencies. To address this, hybrid architectures are gaining traction, where edge computing nodes perform initial data preprocessing closer to sources before forwarding data to the centralized system [31], [70]. Cloud-based centralized DM environments are also being leveraged by DT platforms to improve scalability and availability [31], [32]. Overall, centralized DM provides a holistic and governance-focused foundation for DTs but needs to evolve to be more scalable, distributed, and interoperable to support complex industrial and urban-scale deployments.

B. Distributed Architecture

While centralized architectures have their merits, the massive scale, geographic distribution, and decentralized nature of many DT deployments necessitate more distributed approaches to DM. Distributed DM architectures aim to bring data processing and analytics capabilities closer to the sources of data generation rather than relying on a central repository. In the context of industrial DTs, this often translates to deploying DM capabilities at the edge or on-premise, in close proximity to operational technology (OT) systems, machines, and sensor networks [29], [31], [70] as shown in Fig. 3. Edge computing platforms and gateways can perform local data ingestion, integration, processing, and even analytics before transmitting processed data to higher-level systems or the cloud. This distributed paradigm offers several advantages, including reduced network bandwidth requirements, lower latencies for real-time operations, improved system resilience, data sovereignty, and the ability to enforce localized data governance policies [31], [70]. It also aligns well with the principles of industrial data spaces and data ecosystems, where data ownership and sharing are decentralized [58]. In the context of smart city DTs, distributed architectures can involve deploying DM capabilities across different urban infrastructure domains (e.g., transportation, utilities, buildings) and integrating them through a federated data fabric [13]. This allows each domain to manage its data locally while enabling cross-domain data sharing and integration. Similarly, for healthcare DTs, patient data may be distributed across different medical facilities, necessitating federated DM approaches that respect data sovereignty and privacy regulations while enabling collaborative analytics [21].

Distributed DM architectures often leverage technologies like data virtualization, data fabrics, semantic data lakes, and knowledge graphs to provide a unified view and query interface over disparate, distributed data sources [40], [41], [65]. However, challenges arise in areas like data consistency, schema evolution, and managing complex data flows across distributed nodes. Frameworks like GAIA-X, the Industrial Internet Reference Architecture (IIRA), and the International Data Spaces (IDS) provide guidelines and reference architectures for distributed DM in industrial and IoT contexts, which can be leveraged for DTs [47], [49]. While distributed approaches offer scalability and flexibility, they also introduce additional complexities in areas like data governance, security, and system orchestration. Hybrid architectures that harmoniously combine centralized and distributed elements are emerging as a balanced solution for large-scale DT deployments [31].

C. Hybrid Architecture

Hybrid DM approaches for DTs have emerged to harness the benefits of both centralized and distributed architectures while mitigating their individual limitations. Hybrid architectures synergistically combine centralized repositories with distributed data processing and management capabilities. A common pattern involves deploying lightweight edge nodes or gateways close to data sources for ingestion, preprocessing, filtering, and initial analytics [29]. These edge nodes can buffer and batch data transfers to reduce bandwidth needs. The preprocessed data is then forwarded to a central data hub or cloud platform for further integration, governance, complex analytics, and visualization [31], [58], [65], [70]. Such hybrid architectures allow for localized real-time data processing and decision-making at the edge while still maintaining a consolidated view and historical data storage in the centralized system [31], [32]. This approach strikes a balance between the low latencies and resilience of distributed systems and the governance, scalability, and holistic analytics capabilities of a central DM environment. In industrial contexts, hybrid architectures align with concepts like the Purdue model for control hierarchies and the RAMI4.0 reference architecture, where edge/shop-floor systems interface with higher-level enterprise systems [26], [31]. Commercial industrial IoT platforms like Siemens MindSphere and PTC Thingworx increasingly support hybrid integration of edge and cloud DM. For smart city DTs, a hybrid multi-tier architecture can involve edge nodes for individual infrastructure domains (e.g., buildings, transportation) coupled with district/city-level integration hubs [13], [14]. This federated yet integrated DM approach respects data ownership boundaries while enabling cross-domain insights.

Key challenges with hybrid architectures include managing heterogeneous data flows across distributed and centralized tiers, ensuring data consistency and availability, and optimizing locations for processing and analytics workloads [31]. Emerging technologies like data meshes, data fabrics, and distributed query engines could facilitate more seamless hybrid DM for DTs in the future [40]. Human-centric healthcare scenarios exemplify the applicability of hybrid frameworks, integrating patient data from medical IoT devices and hospital systems via edge analytics with consolidated views in regulatory-compliant cloud data environments [21]. Overall, hybrid DM architectures leverage the strengths of centralized and distributed paradigms, providing a balanced approach to handle the scale, real-time needs, and stakeholder complexity associated with industrial and urban DT deployments.

D. Cloud-Based Data Management

Cloud computing platforms are increasingly being leveraged for centralized DM in DT deployments, capitalizing on the scalability, elasticity, and global accessibility offered by cloud environments [32], [71], [72]. Cloud DM architectures involve provisioning and integrating cloud-native data services like data lakes, data warehouses, stream processing engines, and analytics tools to support the DT data pipeline. The on-demand, pay-as-you-go model of cloud computing enables cost-effective scaling of data storage and processing capabilities in response to fluctuating DT workloads. This elasticity is particularly valuable for DTs in domains like smart cities, where data volumes can surge during events or incidents [14], [71]. Additionally, the global availability and geo-distribution of cloud data centers facilitate low-latency data ingestion and access from distributed geographic sources, which is crucial for DTs spanning large industrial facilities, construction sites, or urban infrastructures [31], [32], [71]. Major cloud providers like AWS, Microsoft Azure, and Google Cloud Platform offer specialized cloud DM services tailored for IoT, time-series data, and DT scenarios [32]. These cloud data platforms often integrate capabilities for data ingestion, storage, integration, processing, analytics, visualization, and machine learning in a cohesive environment.

However, adopting cloud DM for DTs raises concerns around data sovereignty, privacy, vendor lock-in, and the overhead of data transfers over networks [31]. Hybrid models involving cloud integration with on-premises systems or edge processing can help mitigate some of these challenges [76]. Open standards and frameworks promoting interoperability and portability of DT data across cloud environments are crucial to avoid vendor lock-in [77]. Efforts like the Open Industrial Data Hub (OIDH) and the Industrial DT Association (IDTA) are working towards standardizing cloud-based DM for industrial DTs [40]. As cloud DM for DTs continues to evolve, emerging paradigms like data meshes, data fabrics, and distributed cloud architectures could further enhance capabilities for federated yet integrated DM across hybrid cloud/on-premises environments.

E. Blockchain-Based Data Management

Blockchain technology, with its decentralized, immutable, and secure data-sharing capabilities, has garnered interest for DM in DT ecosystems involving multiple stakeholders and untrusted environments [7], [45], [46], [65]. Blockchain-based architectures can facilitate trusted data exchange, provenance tracking, and secure integration of DT data across organizational boundaries [78]. In a blockchain-based DT DM approach, data from disparate sources is recorded as transactions on a distributed ledger maintained by a decentralized peer-to-peer network. This provides transparency, immutability, and tamper-resistance to the data shared across the DT ecosystem [46], [65], [66]. Smart contracts can automate data access control and sharing policies [78]. Potential benefits of adopting blockchain include enhanced data traceability, integrity, security, and trust between parties, which is valuable for multi-stakeholder DT deployments spanning supply chains, construction projects, or smart cities [7], [45], [79]. However, challenges exist regarding the scalability, throughput, standardization, and interoperability of blockchain platforms for industrial DT scenarios [61]. Frameworks like the Ethereum-based EtherTwin combine blockchain, distributed file systems, and Semantic Web technologies to create a secure, decentralized DM fabric for DTs that respects data ownership and sovereignty [46]. SEDIT leverages knowledge graphs along with blockchain to integrate industrial IoT data securely for DTs [57]. While blockchain is still an emerging area for DT DM, its key value propositions lie in trustworthy data integration across multi-party DT ecosystems, secure access control, and data provenance capabilities [45], [66], [78]. Hybrid models combining blockchain with centralized/cloud environments are also being explored [7], [61]. As blockchain scalability improves and standardization efforts crystalize for industrial use cases, blockchain-based approaches could increasingly complement other centralized and distributed DM architectures for secure, multi-party collaboration in DT deployments.

V. Data Management Techniques and Technologies

Effective DM for DTs requires a comprehensive set of techniques and technologies spanning various aspects of the data lifecycle, from data modeling and integration to quality management, security, and visualization. This section explores some of the key techniques and technologies that form the building blocks for robust DM solutions in DT environments. We begin by examining data modeling approaches, including the use of ontologies and semantic models, which play a crucial role in representing and integrating the diverse and heterogeneous data streams that DTs ingest. We discuss techniques for data fusion, entity resolution, and semantic integration, as well as the role of emerging technologies like knowledge graphs and linked data in facilitating data interoperability. Data quality is a critical aspect of DT DM, as the accuracy and reliability of insights derived from the virtual models heavily depend on the quality of the input data. We explore various data quality management techniques, including data profiling, cleansing, deduplication, and outlier detection, as well as methods for monitoring and mitigating data drift over time. Given the sensitive nature of data involved in DTs, data security and access control mechanisms are essential. We delve into techniques like encryption, anonymization, differential privacy, and blockchain-based approaches for ensuring data security and privacy while enabling secure data sharing and collaboration among multiple stakeholders. Data visualization and representation play a vital role in making the complex, multi-dimensional data generated by DTs interpretable and actionable for human stakeholders. We discuss various visualization techniques, including interactive dashboards, immersive technologies like augmented and virtual reality, and approaches for integrating DT data with physical environments. Throughout this section, we highlight the strengths and limitations of different techniques and technologies, as well as their applicability to various DT use cases across domains such as manufacturing, smart cities, healthcare, and others. We also explore emerging trends and research directions in areas like cognitive DM, self-healing data ecosystems, and the integration of DT data with artificial intelligence and machine learning capabilities.

A. Data Modeling and Ontologies

Data modeling and the development of ontologies play a crucial role in enabling effective DM for DTs. Given the extreme heterogeneity of data sources, formats, and semantic models involved, having a well-defined and standardized way to represent and integrate this diverse data is essential [33]. At the core of data modeling for DTs is the concept of ontologies - formal representations of the concepts, entities, properties, and relationships within a particular domain or application [40], [41], [49]. Ontologies provide a shared understanding and semantic foundation for integrating and reasoning over disparate data sources, making them invaluable for DT environments.

Several domain-specific ontologies and data models have been developed and proposed for various DT use cases. For instance, in manufacturing and product lifecycle management, standards like STEP, MIMOSA, and ISA-95 form the basis for modeling product data, asset information, and manufacturing operations data, respectively [39], [47], [49]. The Industry Ontology Foundry (IOF) provides a suite of modular ontologies for industrial DTs [60]. In the construction and building management domains, ontologies based on open standards like IFC, CityGML, and SAREF aim to model BIMs, geospatial data, and sensor data respectively for infrastructure DTs [8], [55], [75]. The W3C Linked Building Data group develops ontologies for integrating building data. Cross-domain ontologies and upper ontologies like DOLCE, BFO, and BORO provide a more generic foundation for integrating multi-domain data required for complex DTs like smart city replicas [40], [41], [80]. Knowledge graphs and linked data approaches based on Semantic Web technologies like RDF, OWL, and SPARQL enable interconnecting these domain ontologies.

However, challenges remain in areas like ontology alignment, mapping, and interoperability across different standards and domains [39], [40], [77]. Techniques like ontology learning, neural ontology embedding, and ontology modularization show promise in semi-automating ontology development and integration tasks [40]. Overall, robust data modeling through well-designed ontologies is crucial for providing a semantic backbone for DT data integration, querying, reasoning, and knowledge extraction. Continued research into developing domain ontologies, cross-domain integration approaches, and ontology engineering best practices is vital for holistic DT DM.

B. Data Fusion and Integration

At the heart of any DT lies the ability to seamlessly fuse and integrate data streams from a myriad of heterogeneous sources, formats, modalities, and semantic models. Effective data fusion and integration techniques are, therefore, critical enablers for creating accurate, up-to-date virtual representations of the physical counterparts.

1) Data Integration: A fundamental challenge in data integration for DTs is dealing with syntactic heterogeneities across data sources, which arise due to differences in data formats, schemas, and storage models [23]. Traditional data integration approaches like extract-transform-load (ETL) pipelines, data virtualization, and enterprise information integration (EII) can be leveraged to resolve these syntactic conflicts and provide a unified data access layer [33]. However, DT environments must also grapple with semantic heterogeneities, where the same concepts are represented differently across domains, data sources, and ontological models [39], [40]. Techniques like ontology mapping, schema matching, entity resolution, and semantic annotation are employed to reconcile these semantic conflicts and establish meaningful relationships across disparate data entities [33], [40], [57].

2) Data Fusion: Once the basic syntactic and semantic integration hurdles are overcome, more advanced data fusion techniques can be applied to derive higher-level insights by combining multi-source data. This includes approaches like data merging, object/feature association, state estimation through sensor fusion (e.g., Kalman filters), and information fusion from multi-modal data using machine learning models [58]. Knowledge graphs and linked data principles based on Semantic Web technologies (RDF, OWL, SPARQL) show promise in providing a unifying framework for integrating and querying data from diverse ontological models [40], [41]. Graph neural networks and geometric deep learning methods are being explored for knowledge graph construction and fusion from multi-modal data [57]. However, scaling data fusion pipelines to handle the volume, velocity, and complexity of industrial-scale DT data remains an open challenge [58]. Distributed data integration architectures, streaming data integration engines, and edge computing paradigms are potential solutions [31], [81]. Automated data integration through machine learning and knowledge-based techniques is also an active area of research [40].

As DTs become more pervasive across domains, interoperability standards, canonical data models, and robust data integration frameworks that can harmonize cross-domain data will become increasingly important [47], [48]. Continued innovation in data fusion techniques that can handle multi-modal, noisy, and dynamic data streams will be vital.

C. Data Quality Management

The accuracy and reliability of insights derived from DTs are fundamentally dependent on the quality of the input data streams feeding the virtual models. Robust data quality management is therefore a critical component of any DT DM strategy. Data quality issues in DTs can arise from various factors - sensor failures, network outages, human errors, data drift over time, conflicting data across sources, and more [23], [30], [43]. These can lead to problems like missing data, duplicate records, outliers/anomalies, inconsistent representations, and integrity violations that undermine the trustworthiness of the twin [29]. Data quality management for DTs involves a range of techniques applied across different stages of the data lifecycle:

- *Data profiling and quality assessment* to detect and quantify quality issues through statistical methods, constraint validation, and data quality rules [23].
- *Data cleansing routines* to handle missing values through interpolation, remove duplicates, smooth noisy data, and identify/correct outliers [23], [43].
- *Data standardization and transformation* to harmonize data representations across sources and resolve syntactic/semantic conflicts [39], [40].
- *Master data and metadata management* to establish authoritative data sources, data lineage tracking, and governance [40], [49].
- *Continuous data quality monitoring* through statistical process control and data drift detection using techniques like concept drift analysis [30], [43], [58].
- *Automated data quality tools* leveraging machine learning for intelligent error detection, data repair, and predictive data quality management [59], [60], [61].
- *Blockchain-based approaches* for ensuring data integrity, traceability, and non-repudiation of data quality metrics [7], [65], [66].

Data quality scorecards, key performance indicators (KPIs), stewardship processes, and organizational accountability are also essential governance aspects [40], [49]. Reinforcement learning and agent-based frameworks show promise in developing autonomous, self-healing data quality management capabilities for DTs [59], [60], [61]. As DTs integrate more diverse, real-time data streams, data quality management becomes more challenging but also more critical. Techniques that can handle multi-modal data quality issues in a scalable, automated fashion while quantifying and propagating uncertainties will be vital [5], [27]. Ultimately, a holistic data quality strategy covering people, processes, and technology dimensions is key to ensuring trustworthy DTs that drive reliable decision-making.

D. Data Security and Access Control

DTs often deal with sensitive data related to physical assets, operational processes, intellectual property, and even personal information about individuals. Ensuring the security and privacy of this data is paramount, necessitating robust security controls and access management mechanisms throughout the data lifecycle.

1) Data Security: Data security for DTs starts with secure data acquisition and transmission from source systems and IoT devices. Techniques like encryption (e.g., AES, RSA), secure communication protocols (TLS/SSL), and device authentication/authorization frameworks are employed to protect data in transit [29], [31], [45]. Blockchain-based approaches for device identity and data provenance management are also being explored [7], [65], [66]. For data at rest, encryption, tokenization, and masking techniques help protect sensitive data stored in databases, data lakes, and other repositories [31], [45]. Secure storage solutions like confidential computing, hardware security modules (HSMs), and encrypted filesystems can further enhance protection.

2) Access Control: Access control mechanisms govern who can access which DT data assets and what operations they can perform. Role-based access control (RBAC) and attribute-based access control (ABAC) models are commonly used and enforced through authentication, authorization, and auditing controls [31], [45], [46]. Federated identity management and secure access gateways facilitate controlled data sharing across organizational boundaries. Data anonymization and privacy-preserving techniques like differential privacy, secure multi-party computation, and federated learning enable analyzing and sharing insights from sensitive data without exposing raw data [45], [81]. However, finding the right balance between privacy, security, and data utility remains an open challenge.

From a systems perspective, secure data enclaves, micro-segmentation, zero-trust architectures, and security information and event management (SIEM) solutions are critical for comprehensive data security in DT environments [31], [45]. Continuous security monitoring, vulnerability assessments, and incident response processes are also essential. Blockchain and distributed ledger technologies (DLTs) have been proposed as enablers for secure, decentralized data sharing and trusted audit trails in multi-party DT contexts [7], [46], [65], [66]. However, their integration with existing security controls and scalability for industrial DTs needs further research. As DTs become more interconnected and data sharing increases, a holistic "security by design" mindset coupled with robust governance frameworks and regulatory compliance will be vital to protect this critical data infrastructure.

E. Data Visualization and Representation

DTs generate massive volumes of multidimensional, multi-modal data that need to be effectively visualized and represented for human comprehension, analysis, and decision support. Intuitive data visualization and innovative representation techniques play a vital role in translating complex virtual models into actionable insights. At a fundamental level, DT visualization relies on traditional techniques like dashboards, charts, graphs, and reports to present KPIs, trends, and aggregated views derived from sensor data, system logs, and other operational data sources [16], [25], [50], [80]. Interactive features like filtering, drill-downs, and what-if analysis capabilities enhance the utility of these visualizations.

However, DTs often require more advanced 3D visualization and integration with actual physical environments and assets. Techniques like 3D modeling, CAD integration, BIM, geospatial information systems (GIS), and physics-based rendering are employed to create immersive visual replicas [8], [11], [18], [55]. Augmented reality (AR) and mixed reality (MR) technologies that overlay DT data and visualizations onto physical environments are gaining traction, enabling new ways to visualize, interact with, and control cyber-physical systems [25], [50]. Similar immersive experiences are possible through virtual reality (VR) environments that allow users to explore and interact with DTs in a fully virtualized space. Beyond visual representations, techniques like sonification (data-driven audio), haptics (data-driven touch feedback), and other multi-sensory interfaces are being explored to enhance the experience of interacting with DTs, particularly in training, maintenance, and design review applications [82].

As the complexity and scale of DTs increase, challenges arise in areas like real-time visualization of massive data streams, multi-user collaborative visualization, and cross-platform/cross-domain visualization interoperability [25], [50]. Distributed visualization architectures, edge computing paradigms, and web-based visualization frameworks are potential solutions to these scalability and performance issues. Overall, data visualization and representation play a crucial role in deriving value from DTs by making the complex data generated by these systems comprehensible and actionable for human stakeholders. Continued innovation in multi-modal visualization, immersive technologies, and scalable visualization architectures will be vital as DTs become more pervasive.

VI. Data Management for Digital Twin Applications

DM considerations and approaches for DTs vary significantly across different application domains [11], [18], [19], [20], [21], [22]. Each domain presents unique challenges stemming from the diverse nature of data sources, formats, semantics, and stakeholders involved. Robust DM strategies tailored to the specific domain requirements are crucial for realizing the full potential of DTs. This section explores the DM landscape across various domains where DTs are being deployed, such as manufacturing, smart cities, construction, healthcare, energy, transportation, and agriculture [11], [18], [19], [20], [21], [22]. It examines the distinct DM challenges posed by each domain, as well as the architectural approaches, techniques, and best practices employed to address these challenges effectively. Table II summarizes the key DM challenges and solutions across different domains. The manufacturing and Industry 4.0 contexts require seamless integration of operational data from industrial assets, sensors, and control systems with design and simulation data from product lifecycle management tools [24], [26]. Smart city DTs must grapple with the complexities of harmonizing heterogeneous data formats, models, and semantics across multiple urban domains [13], [14]. In construction and building management, the focus is on integrating BIMs with real-time sensor data and enabling data traceability across the construction lifecycle [8], [11], [55], [75]. Healthcare and human DTs involve managing sensitive personal data from diverse sources like medical records, biosensor data, and genomic data while ensuring data privacy and security [21]. Other domains such as energy, transportation, and agriculture face challenges in processing real-time sensor data from distributed assets, integrating with external data contexts, and enabling data-driven simulations and predictions [22], [83].

A. Manufacturing and Industry 4.0

DTs are increasingly being adopted in manufacturing and Industry 4.0 environments to create virtual representations of production systems, processes, and products [1], [3], [4], [24], [26]. These DTs rely on the effective integration and management of diverse data streams from industrial assets, sensors, control systems, enterprise systems, simulation tools, and product lifecycle management applications [24], [28], [41], [42], [88]. A key DM challenge in this domain is the need to harmonize and fuse real-time operational data from the shop floor with design and simulation data represented using standards like CAD, FEM, and other modeling paradigms [39], [47], [49], [89]. Bridging the semantic gap between as-designed and as-operated data is crucial for creating accurate DT models that can enable monitoring, diagnostics, and optimization capabilities. Another

TABLE II
KEY CHALLENGES AND SOLUTIONS ACROSS DOMAINS

Domain	Key Data Management Challenges	Solutions & Technologies
Manufacturing [1], [3], [4], [24], [26]	• Harmonizing real-time operational data with design/simulation data • Ensuring data quality, traceability, and provenance • Secure data sharing across the supply chain • Real-time processing at industrial scale	• Edge-Cloud architectures • Distributed data architectures • Blockchain for data sovereignty • Deep learning algorithms • Cloud-based solutions
Smart Cities [13], [14]	• Extreme heterogeneity of data formats and models • Massive scale and velocity of data • Data security and privacy concerns • Multi-stakeholder collaboration • Cross-domain integration	• Distributed data architectures • Semantic web technologies • Knowledge graphs • Cross-domain ontologies • Cloud and edge computing • Blockchain
Construction [8], [11], [55], [75]	• Integration of BIM with real-time sensor data • Handling geospatial and 3D data • Data traceability across asset lifecycle • Multi-stakeholder data sharing • Version control and change management	• Semantic web and linked data approaches • Distributed ledger technologies • Cloud-based DM platforms • BIM collaboration tools
Healthcare [21], [60], [84]	• Data privacy and security • Multi-modal data integration • Data quality and trustworthiness • Cross-institutional data sharing • Complex data heterogeneity	• Privacy-preserving techniques • Federated learning • Differential privacy • Comprehensive ontologies • Knowledge graphs • Federated data ecosystems
Energy [83], [85]	• Massive scale of real-time operational data • Data quality and trustworthiness • Security of critical infrastructure data • Cross-system integration • Real-time processing requirements	• Cloud computing • Edge computing architectures • Modular design approaches • AI-enhanced modeling • Real-time simulators
Transportation [56], [59], [86], [87]	• Managing trajectory data from mobile assets • Real-time processing requirements • Data quality and privacy • Cross-system integration • Multi-stakeholder collaboration	• Specialized trajectory data management • Standardized data models • Data exchange protocols • LSTM models • DEA models
Agriculture [22]	• IoT big data management (13 V's) • Integration with external contextual data • Secure stakeholder data sharing • Real-time monitoring requirements • Data quality and traceability	• Edge computing • Distributed processing architectures • Hybrid cloud-edge systems • Secure data sharing protocols • IoT data management platforms

critical aspect is ensuring data quality, traceability, and provenance across the entire manufacturing data lifecycle [30], [42], [43], [90]. Robust data governance mechanisms, including data cleansing, master DM, and metadata tracking, are essential to maintain the trustworthiness and reliability of insights derived from DTs. Quantifying and propagating uncertainties inherent in the data is also vital [5], [27], [63].

Enabling secure and controlled data sharing across the extended supply chain and manufacturing ecosystem is another key requirement [31], [45], [46]. DTs often involve collaboration between multiple stakeholders, necessitating robust access control, data sovereignty, and intellectual property protection mechanisms. Distributed ledger and blockchain technologies are being explored as potential solutions [7], [46], [65], [66], [91]. Real-time data processing and analytics at the scale and velocity demanded by industrial DTs pose significant scalability and performance challenges [32], [58]. Edge computing paradigms, distributed data architectures, and cloud-based solutions are being leveraged to address these demands [31], [32], [70], [81], [92].

Wu et al. [89] presents a Digital Twin (DT) framework for intelligent small surface defect detection in cyber-manufacturing systems as shown in Fig. 4. This framework integrates physical and digital components to enable real-time monitoring, analysis, and control of surface defect detection processes. The proposed system leverages an Edge-Cloud architecture to efficiently collect, process, and analyze multi-modal data (imaging and depth data) produced by factory sensors. Edge nodes preprocess the data before transmitting it to the cloud for further analysis. A deep learning-based algorithm is employed to detect surface defects with high accuracy and recall, thus constructing a DT in the cyber-manufacturing environment. The proposed framework demonstrates the potential of DTs in enhancing small surface defect detection tasks in manufacturing settings.

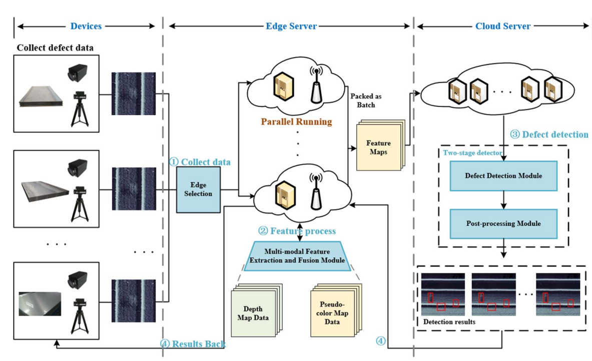

Fig. 4. Digital twin framework for intelligent small surface defect detection in cyber-manufacturing systems [89].

B. Smart Cities and Urban Planning

DTs are emerging as powerful tools for urban planning, city management, and the development of smart city solutions [13], [14]. These virtual replicas consolidate and integrate data from multiple domains within a city's ecosystem, such as transportation networks, utility grids, buildings, public services, and citizen data sources [22], [80]. One of the primary DM challenges in smart city DTs is dealing with the extreme heterogeneity of data formats, models, and semantics across these diverse urban domains [40], [41]. Providing semantic integration and interoperability across IoT data streams, GIS, BIMs, and other city data sources is a complex undertaking [33]. The massive scale and velocity of data involved in city-scale DTs pose significant scalability and performance demands [32], [58]. Ingesting, processing, and analyzing real-time data streams from thousands of sensors and IoT devices in a timely manner is crucial for enabling effective monitoring, simulation, and decision-support capabilities. Ensuring data security and privacy is another critical consideration, as smart city DTs often involve sensitive data related to infrastructure, operations, and citizen activities [31], [45]. Robust cybersecurity measures, access controls, and privacy-preserving DM techniques are essential to protect this critical data infrastructure [44], [93]. Data governance and stakeholder collaboration are also key aspects, as smart city DTs involve multiple agencies, utilities, and stakeholders [31], [46]. Enabling secure and controlled data sharing, distributed governance models, and establishing clear data ownership and sovereignty mechanisms are vital [77]. To address these multifaceted challenges, smart city DT initiatives are leveraging technologies like distributed data architectures, cloud computing, edge computing, and blockchain [31], [32], [45], [81], [93]. Semantic web technologies, knowledge graphs, and cross-domain ontologies are also being explored for data integration and interoperability [33], [40], [41]. Ultimately, robust DM frameworks that can harmonize the diverse data landscape, ensure scalability and performance, and enable secure and collaborative data ecosystems are critical for unlocking the full potential of DTs in urban planning and smart city applications.

Schrotter and Hürzeler [14] present the digital twin of the City of Zurich as a tool for urban planning and decision-making. The digital twin integrates 3D spatial data and models of various urban themes, such as buildings, bridges, and vegetation, to create a comprehensive digital representation of the city. This digital twin serves as a valuable data foundation for urban planning processes, enabling visualization, analysis, and simulation of different scenarios. By releasing 3D spatial data under Open Government Data initiatives, the City of Zurich promotes public awareness, fosters collaboration, and simplifies the development of applications that leverage the digital twin. The digital twin facilitates the simulation of urban climate scenarios, linking results to existing 3D spatial data and supporting decision-making processes in a comprehensible manner. The lifecycle management of the digital twin, including the description of 3D spatial data, models, and metadata, is crucial for ensuring the accuracy and relevance of the digital representation.

C. Construction and Building Management

DTs are increasingly being adopted in the construction and building management sectors to create virtual representations of infrastructure assets and facilities [8], [11], [55], [75]. These DTs leverage data from multiple sources, including BIMs, sensor networks deployed in buildings and construction sites, as well as facility management systems and operational data stores [8], [55], [75], [80]. A key DM challenge in this domain is the integration of BIM data with real-time sensor data streams from the physical assets [8], [55], [75]. BIMs, which capture the design and as-built information of infrastructure assets, are typically represented using standards like Industry Foundation Classes (IFC) and CityGML [8], [75]. Harmonizing the semantics and data models of BIMs with the diverse sensor data formats and representations poses significant interoperability hurdles [33]. Handling the geospatial and 3D nature of construction and building data is another important consideration [11], [55]. Effective management of geospatial data, 3D models, and point cloud data from sources like laser scanning and photogrammetry is crucial for creating accurate DT representations [80]. Data traceability and provenance across the asset lifecycle, from design and construction to operations and maintenance, is vital for ensuring data integrity and enabling data-driven decision-making [8], [11], [63]. Robust data governance mechanisms, including version control, change management, and auditing capabilities, are essential. Enabling secure and controlled data sharing among multiple stakeholders, such as designers, contractors, facility managers, and building owners, is another key requirement [31], [46]. Access control, data sovereignty, and intellectual property protection mechanisms play a critical role in facilitating collaborative DT environments [77]. To address these challenges, construction and building management DTs are leveraging technologies like Semantic Web and linked data approaches for data integration [40], [55], [75], as well as distributed ledger technologies for ensuring data provenance and secure data sharing [7], [33], [46], [65], [66]. Cloud-based DM platforms and BIM collaboration tools are also being widely adopted [31], [32], [55].

Lu et al. [11] present a case study of developing a digital twin at building and city levels, focusing on the West Cambridge campus. The digital twin integrates heterogeneous data sources, supports effective data querying and analysis, and facilitates decision-making processes in operation and maintenance (O&M) management. The proposed system architecture for digital twins at both building and city levels as shown in Fig. 5 is designed to bridge the gap between human relationships with buildings and cities. The digital twin demonstrator developed for the West Cambridge site of the University of Cambridge in the U.K. serves as a practical example of implementing digital twins in real-world scenarios. The study aims to provide a clear roadmap, share lessons learned, and address challenges involved in developing digital twins for building and city levels, offering valuable insights for asset management practitioners, policymakers, and researchers.

D. Healthcare and Human Digital Twins

The concept of DTs is also gaining traction in the healthcare domain, where virtual representations of human physiology, biology, and health parameters can enable personalized medicine, predictive healthcare, and improved treatment outcomes [21],

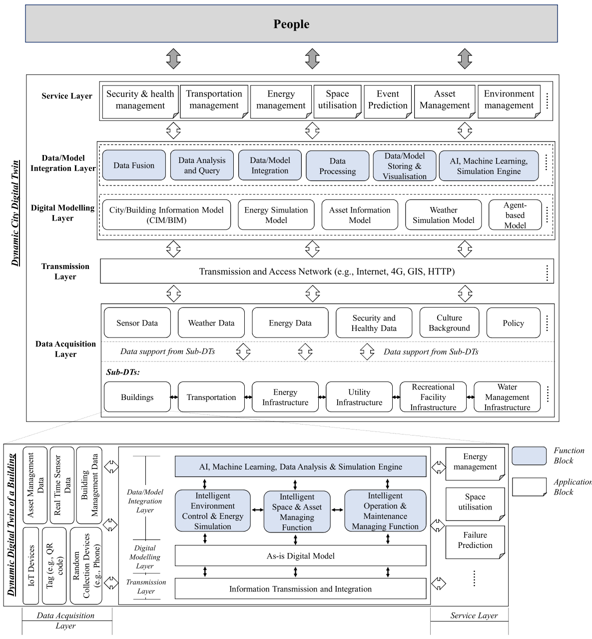

Fig. 5. West cambridge campus digital twin system architecture [11].

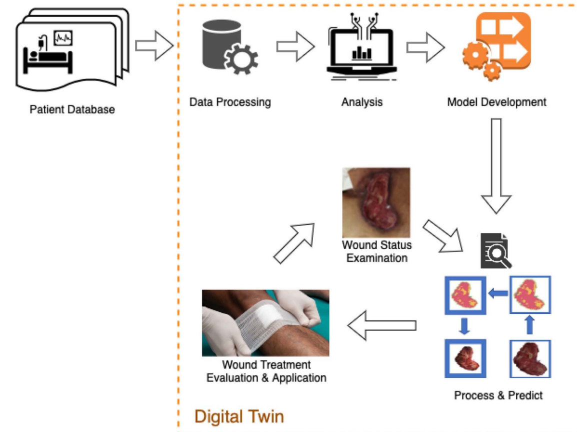

Fig. 6. Digital twin framework for chronic wound management [84].

[84]. These human DTs integrate diverse data sources, including electronic medical records, biosensor data, genomic information, lifestyle data, and biomedical knowledge bases [21], [60].

A critical DM challenge in healthcare DTs is ensuring data privacy and security while enabling cross-institutional data sharing and collaborative analytics [21], [45], [60], [84]. Healthcare data often contains sensitive personal information, necessitating robust privacy-preserving techniques like data anonymization, differential privacy, and federated learning approaches [45], [60], [81]. Implementing these techniques at scale while preserving data utility is a complex undertaking. Dealing with the heterogeneity and complexity of multi-modal healthcare data is another key hurdle [33], [40], [41], [84]. Integrating structured data from electronic health records with unstructured data from medical imaging, genomics, and patient-generated data requires advanced data fusion and integration techniques. Developing comprehensive ontologies and knowledge graphs to represent and reason over this diverse healthcare data landscape is an active area of research [33], [40], [57], [84]. Data quality, reliability, and trustworthiness are paramount in healthcare applications, as insights derived from DTs can directly impact patient care and treatment decisions [23], [30], [43], [84]. Robust data quality assurance processes, uncertainty quantification methods, and rigorous data governance frameworks are essential to ensure the trustworthiness of healthcare DTs. From an architectural perspective, distributed DM approaches that respect data sovereignty and enable cross-institutional collaboration are gaining traction in healthcare [31], [60], [70], [81]. Federated data ecosystems and secure multi-party computation techniques show promise in addressing privacy and data-sharing challenges [45], [81].

Sarp et al. [84] propose the implementation of a digital twin in healthcare for chronic wound management as shown in Fig. 6. Leveraging artificial intelligence techniques, the digital twin is composed of data collection, data processing, and AI models dedicated to wound healing. A novel AI pipeline is utilized to track the healing of chronic wounds, with the digital twin simulating and replicating the healing process. The proposed wound-healing prediction model guides the treatment of chronic wounds, enabling early identification of non-healing wounds and facilitating timely adjustments to the treatment plan. By incorporating a digital twin in healthcare, the proposed system enables personalized and tailored treatments, potentially playing a crucial role in proactive problem identification.

E. Energy

Digital twins are increasingly being deployed across the energy sector to create virtual replicas of power generation assets, distribution networks, and energy storage systems [83], [85]. In the power transmission domain, DTs enable monitoring and optimization of complex systems like UHVDC (Ultra High Voltage Direct Current) transmission, where accurate loss measurement and prediction are critical [85]. These energy sector DTs must effectively integrate and manage data from diverse sources including sensor networks, control systems, weather data, and energy demand forecasts [83].

A key data management challenge in energy DTs is handling the massive scale and velocity of real-time operational data streams from distributed energy assets [58], [88]. Power systems generate continuous streams of measurement data that must be processed in real-time to enable effective monitoring, control, and optimization. This requires robust data architectures that can support high-throughput data ingestion and processing while maintaining data quality and reliability [92]. Data quality and trustworthiness are particularly critical in energy DTs, as decisions based on DT insights can directly impact grid stability and reliability [88]. Robust data validation, cleaning, and uncertainty quantification methods are essential. Additionally, the energy sector must deal with challenges around data security and privacy, as power system data is often considered critical infrastructure information [83], [88]. The domain also faces

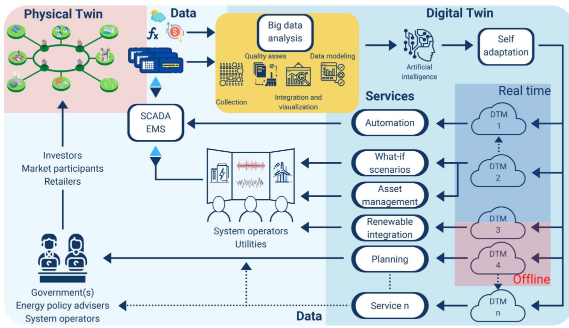

Fig. 7. Modular design of a power system digital twin [92].

unique challenges around data integration and interoperability across different energy subsystems and stakeholders [83]. Standards and protocols for data exchange, along with semantic integration approaches, are becoming increasingly important for enabling effective collaboration and data sharing in energy DT implementations [77].

To address these challenges, energy sector DTs are leveraging technologies like cloud computing and edge computing architectures [72]. Cloud platforms provide the scalability and computational resources needed for complex analytics and simulations, while edge computing enables local processing of time-sensitive data streams [72]. Advanced modeling techniques, including AI-enhanced approaches, are being explored to improve the accuracy and capabilities of energy DTs [88]. Felipe Arraño-Vargas and Georgios Konstantinou [92] propose a modular design of a power system digital twin as shown in Fig. 7. The modular design approach decomposes complex energy systems into modular components with well-defined interfaces, enabling better management of data integration challenges and more flexible, maintainable DT implementations. This modularity also supports the integration of real-time simulators, which are essential for power system analysis and optimization. The proposed modular design framework provides a structured approach to developing power system DTs, facilitating the integration of diverse data sources and enabling scalable, extensible DT implementations.

F. Transportation

Digital twins in the transportation sector are being developed to create virtual representations of vehicles, fleets, transportation infrastructure, and entire mobility systems [56], [59], [86], [87]. These DTs support various applications including traffic management, infrastructure monitoring, and transportation system optimization. The transportation domain presents unique data management challenges due to its dynamic nature and the need to integrate data from mobile assets with fixed infrastructure systems.

A significant data management challenge in transportation DTs is handling trajectory data from moving vehicles and mobile assets [58]. This requires specialized approaches for managing, analyzing, and learning from trajectory datasets while dealing with issues like data quality, privacy, and real-time processing requirements [58]. The integration of trajectory data with other contextual information, such as weather conditions and infrastructure status, adds additional complexity to the data management challenge. Transportation DTs must also address challenges related to data interoperability and integration across different transportation subsystems and stakeholders [77]. This includes harmonizing data formats and semantics between various transportation modes, infrastructure components, and operational systems. The development of standardized data models and exchange protocols is crucial for enabling effective collaboration and data sharing in transportation DT implementations [77].

Infrastructure efficiency assessment represents a key application area for transportation DTs. Tu et al. [87] propose a Digital Twins-based automated pilot for energy-efficiency assessment of intelligent transportation infrastructure. The study establishes a Data Envelopment Analysis (DEA) model for transportation infrastructure efficiency evaluation under Digital Twins technology. Using the transportation infrastructure of 12 prefecture-level cities in Jiangsu Province as the research object, the Digital Twins DEA model and the traditional Stochastic Frontier Approach (SFA) model are employed to estimate the efficiency of transportation infrastructure in these cities. The study also simulates and predicts traffic flow data of a road section in Zhenjiang City using the Long Short-term Memory (LSTM) traffic flow prediction model. The results demonstrate the effectiveness of the Digital Twins DEA model in calculating transportation infrastructure efficiency and the accuracy of the LSTM-based traffic flow prediction model.

G. Agriculture and Precision Farming

Digital twins are finding increasing application in agriculture and precision farming, enabling virtual representations of farms, crop fields, livestock, and agricultural machinery [22]. These agricultural DTs support data-driven decision making by integrating diverse data sources including IoT sensors, drone imagery, satellite data, weather information, and crop modeling tools. The agricultural domain presents unique data management challenges due to the biological nature of the systems being monitored and the strong influence of environmental factors.

The management of IoT Big Data presents a significant challenge in agricultural DTs [29]. The proliferation of IoT sensors in modern farming operations generates massive volumes of data across multiple dimensions (the "13 V's" of Big Data), requiring sophisticated approaches for data collection, storage, processing, and analysis. Key challenges include handling data variety (multiple data types and formats), velocity (real-time data streams), and veracity (ensuring data quality and reliability) [29]. Data integration and contextualization are particularly critical in agricultural DTs, as farming operations are heavily influenced by external factors such as weather conditions, soil characteristics, and market dynamics [22]. The data management system must enable effective integration of internal farm data with external contextual information while maintaining data quality and traceability. This includes handling both structured data (e.g., sensor readings, production records) and unstructured data (e.g., imagery, weather forecasts) in a coherent manner.

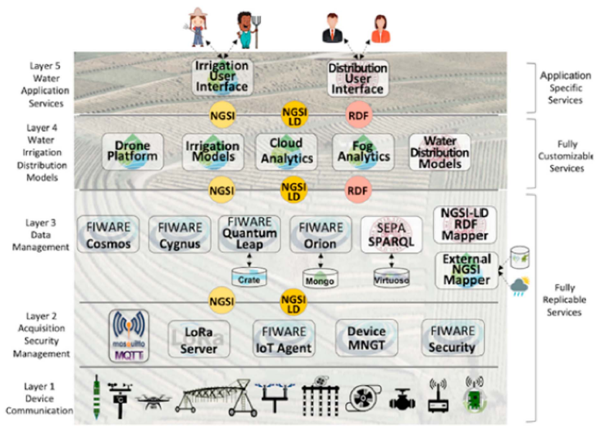

Fig. 8. SWAMP project architecture for smart water management in precision agriculture [22].

Another key consideration is enabling secure and controlled data sharing among various stakeholders in the agricultural value chain [31], [46]. Agricultural DTs often involve collaboration between farmers, agricultural service providers, researchers, and other parties, necessitating robust mechanisms for data access control and privacy protection. Distributed data architectures and secure data sharing protocols play an important role in facilitating these collaborative scenarios while protecting sensitive farm data [31].

Alves et al. [22] present the SWAMP project architecture for smart water management in precision agriculture as shown in Fig. 8. The SWAMP project aims to develop an Internet of Things (IoT) platform for water management in farms, leveraging technologies developed by the Sensing Change project. The proposed digital twin environment integrates soil probes and IoT devices to create a cyber-physical system (CPS) that enables farmers to monitor and manage water resources and equipment on their farms. The system collects data from soil probes and displays it in a dashboard, providing insights into resource usage and farm conditions. The digital twin environment facilitates the deployment of additional monitoring and control devices, enabling a comprehensive digital representation of the farm. The SWAMP project architecture demonstrates the potential of digital twins in precision agriculture for optimizing resource management and enhancing farm productivity.

VII. Open Research Challenges and Future Directions

The rapid development and adoption of DT technology across various domains has brought numerous opportunities as well as challenges. While significant progress has been made in areas such as DM architectures, modeling techniques, and application-specific implementations, several critical research challenges remain to be addressed. This section highlights some of the key open issues and future research directions in the field of DM for DTs. These challenges span topics such as interoperability and standardization, real-time data processing, intelligent and cognitive DM approaches, and ethical and regulatory considerations. Addressing these challenges will be crucial for realizing the full potential of DTs and enabling their widespread and effective deployment across diverse industries and applications.

A. Interoperability and Standardization

Interoperability and standardization are critical challenges that need to be addressed to enable seamless integration and collaboration among different DT systems and their data sources [2]. Currently, there is a lack of widely adopted standards and common data models for representing and exchanging DT data across various domains and applications [33], [77]. This heterogeneity in data formats, communication protocols, and semantic models hinders the interoperability of DTs and limits their ability to share and integrate data from multiple sources effectively [23], [41]. Developing comprehensive and widely accepted standards is essential for ensuring data consistency, compatibility, and interoperability among DT systems [2]. These standards should cover areas such as data exchange formats, ontologies, communication protocols, security mechanisms, and application programming interfaces (APIs) [77]. Standardization efforts should involve collaboration among industry stakeholders, academic researchers, and regulatory bodies to ensure the widespread adoption and implementation of these standards [15], [47]. Furthermore, research is needed to develop flexible and extensible data integration frameworks that can handle the diverse and heterogeneous data sources associated with DTs [23], [39], [57]. These frameworks should support semantic data integration, enabling the mapping and translation of data across different domains and ontologies. They should also incorporate mechanisms for data quality assessment, conflict resolution, and provenance tracking to ensure the reliability and trustworthiness of the integrated data [7], [40]. By addressing the challenges of interoperability and standardization, DT systems can overcome data silos, enabling seamless data sharing, integration, and collaboration across different organizations, domains, and applications [31], [74]. This will unlock new opportunities for data-driven decision-making, optimization, and innovation in various sectors, including manufacturing, healthcare, smart cities, and beyond [4].

B. Real-Time Data Processing and Analytics

Real-time data processing and analytics are essential for enabling DTs to provide up-to-date insights and support timely decision-making. DTs often involve the integration of high-velocity data streams from various sources, such as sensors, IoT devices, and operational systems. Effectively managing and analyzing these data streams in real-time presents significant challenges that need to be addressed [29], [30], [58], [70]. One key challenge is the development of scalable and efficient data ingestion and processing pipelines that can handle the high volume, velocity, and variety of data generated by DT systems. These pipelines should be able to ingest, filter, and preprocess data from multiple heterogeneous sources while ensuring low latency and high throughput [31], [94]. Research is needed to explore distributed and parallel computing techniques, as well as stream processing frameworks, to enable real-time data processing at scale. Another important challenge is the development of advanced real-time analytics and decision-support capabilities for DTs. This includes techniques for real-time data fusion, pattern recognition, anomaly detection, and predictive modeling [43]. These analytics should leverage machine learning, artificial intelligence, and other advanced data analysis methods to extract valuable insights from the continuous data streams and support real-time decision-making processes. Furthermore, research is needed to address the challenges of data quality, reliability, and trustworthiness in real-time data processing scenarios. Techniques for real-time data validation, cleansing, and conflict resolution should be developed to ensure the integrity and accuracy of the data used for decision-making [23], [40]. Additionally, mechanisms for provenance tracking and data lineage management are crucial for maintaining transparency and accountability in real-time analytics processes. Addressing these challenges will enable DTs to provide real-time visibility, monitoring, and control capabilities, unlocking new opportunities for optimizing processes, improving operational efficiency, and enabling proactive maintenance and decision-making across various industries, such as manufacturing, healthcare, and smart cities [26], [83], [95].

C. Intelligent and Cognitive Data Management

As DT systems become increasingly complex, involving vast amounts of heterogeneous data from multiple sources, traditional DM approaches may prove insufficient. There is a growing need for intelligent and cognitive DM techniques that can autonomously and adaptively handle the challenges of data integration, quality assurance, security, and decision support [43]. One key area of research is the development of self-learning and self-optimizing DM systems that can continuously improve their performance based on feedback and experience. These systems should leverage machine learning, artificial intelligence, and knowledge representation techniques to learn patterns, identify anomalies, and adapt to changing data characteristics and requirements [12], [54], [96]. Another important challenge is the integration of cognitive capabilities into DM processes, enabling DTs to reason, learn, and make intelligent decisions based on the available data. This includes techniques for knowledge extraction, inference, and reasoning, as well as explainable AI methods that can provide transparency and interpretability for the decisions made by cognitive DM systems [19], [43]. Furthermore, research is needed to develop adaptive and context-aware DM approaches that can dynamically adjust to changing operational conditions, user preferences, and application requirements. These approaches should leverage techniques such as context modeling, situational awareness, and dynamic adaptation to optimize DM processes and ensure the delivery of relevant, timely, and trustworthy information to support decision-making [19], [31], [94]. Addressing these challenges will enable DTs to transition from static data repositories to intelligent and cognitive systems capable of autonomously managing and interpreting complex data, deriving insights, and supporting decision-making processes. This will unlock new opportunities for enhanced operational efficiency, predictive maintenance, and optimized resource allocation across various domains, such as manufacturing, healthcare, and smart cities [4], [83], [95].

D. Ethical and Regulatory Considerations

As DT technology continues to evolve and become more prevalent across various domains, it is crucial to address the ethical and regulatory considerations surrounding DM and usage. DTs often involve the collection, processing, and integration of sensitive data, including personal information, operational data, and intellectual property. Ensuring the responsible and ethical management of this data is essential for maintaining public trust, mitigating risks, and enabling the sustainable development of DT technologies. One key challenge is the development of robust data governance frameworks and policies that balance the benefits of data-driven insights with the protection of individual privacy and data rights. These frameworks should incorporate principles of data minimization, purpose limitation, and user consent, while also considering the unique challenges posed by DTs, such as data integration and long-term data retention [23], [45], [46]. Another important consideration is the ethical use of data and AI-driven decision-making in DT systems. As these systems become more intelligent and autonomous, there is a need to ensure that their decisions and recommendations are fair, unbiased, and aligned with ethical principles [19]. Research is needed to develop techniques for ethical AI, including explainable AI, bias detection, and value alignment, to promote transparency, accountability, and trust in DT systems. Furthermore, regulatory compliance is a critical issue that must be addressed in the context of DT DM. DTs may be subject to various regulations and standards related to data privacy, cybersecurity, and industry-specific requirements. Research is needed to develop frameworks and tools for continuous monitoring, auditing, and compliance management to ensure that DT systems operate within the relevant legal and regulatory boundaries [40], [45]. Addressing these ethical and regulatory challenges will not only promote the responsible development and deployment of DT technologies but also foster public trust and confidence in these systems. This will be essential for enabling the widespread adoption and acceptance of DTs across various sectors, while mitigating potential risks and ensuring the protection of individual rights and societal values.

VIII. Conclusion

The rapid proliferation of DT technologies across various industries has underscored the critical importance of robust DM solutions. Effective DM lies at the core of realizing the full potential of DTs, enabling seamless integration, processing, and analysis of the diverse and complex data streams that drive these virtual representations. This comprehensive review has explored the multifaceted landscape of DM challenges, architectures, techniques, and applications in DTs. From addressing data heterogeneity and quality issues to ensuring scalability, security, and real-time performance, the paper has highlighted the key hurdles and best practices that shape the DM strategies for DT systems. As DTs continue to evolve and integrate with emerging technologies such as artificial intelligence, edge computing, and blockchain, new frontiers of research and innovation in DM are opening up. Areas such as interoperability and standardization, real-time data processing, intelligent and cognitive DM, and ethical and regulatory considerations present exciting opportunities for researchers and practitioners alike.

Addressing these challenges will be crucial in enabling the widespread adoption and sustainable development of DT technologies across diverse domains, from manufacturing and smart cities to healthcare and precision agriculture. By unlocking the power of data-driven insights and enabling informed decision-making, robust DM solutions will play a pivotal role in optimizing processes, enhancing operational efficiency, and driving innovation in the era of cyber-physical systems. Ultimately, effective DM is not just a technical necessity but a strategic imperative for organizations seeking to harness the transformative potential of DTs. As this review has demonstrated, a holistic and multidisciplinary approach, combining cutting-edge techniques from domains such as data engineering, artificial intelligence, cybersecurity, and regulatory compliance, will be essential in shaping the future of DM for DT systems.

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